HACKATHON PHASE 1: IDEA & PITCH

Q-Vote: Post-Quantum Secure Democracy

A modular, privacy-preserving e-voting system designed to protect universal elections from the quantum threat.

Universal Application: Municipal • Organizational • National

The Quantum Threat

Current e-voting systems (RSA/ECC) are vulnerable to **Shor's Algorithm**. Future quantum computers can break these in polynomial time.

- **Harvest Now, Decrypt Later:** Adversaries are storing traffic today to decrypt votes in the future.
- **Broken Anonymity:** Retroactive decryption destroys the fundamental right to a secret ballot.
- **Institutional Decay:** Loss of trust in digital governance leads to democratic instability.

SDG 16: Strong Institutions

Target 16.6 & 16.7

Our project builds the "Accountable and Transparent Institutions" of the future by ensuring:

Inclusive Participation: Secure, remote voting lowers barriers for marginalized communities.

Public Trust: Mathematical proofs of integrity replace blind trust in centralized authorities.

Scalable Justice: A modular framework applicable to any democratic scale.

The 3-Pillar Framework

PQC Signatures

NIST-approved CRYSTALS-Dilithium ensures that ballot identities are forge-proof even against quantum adversaries.

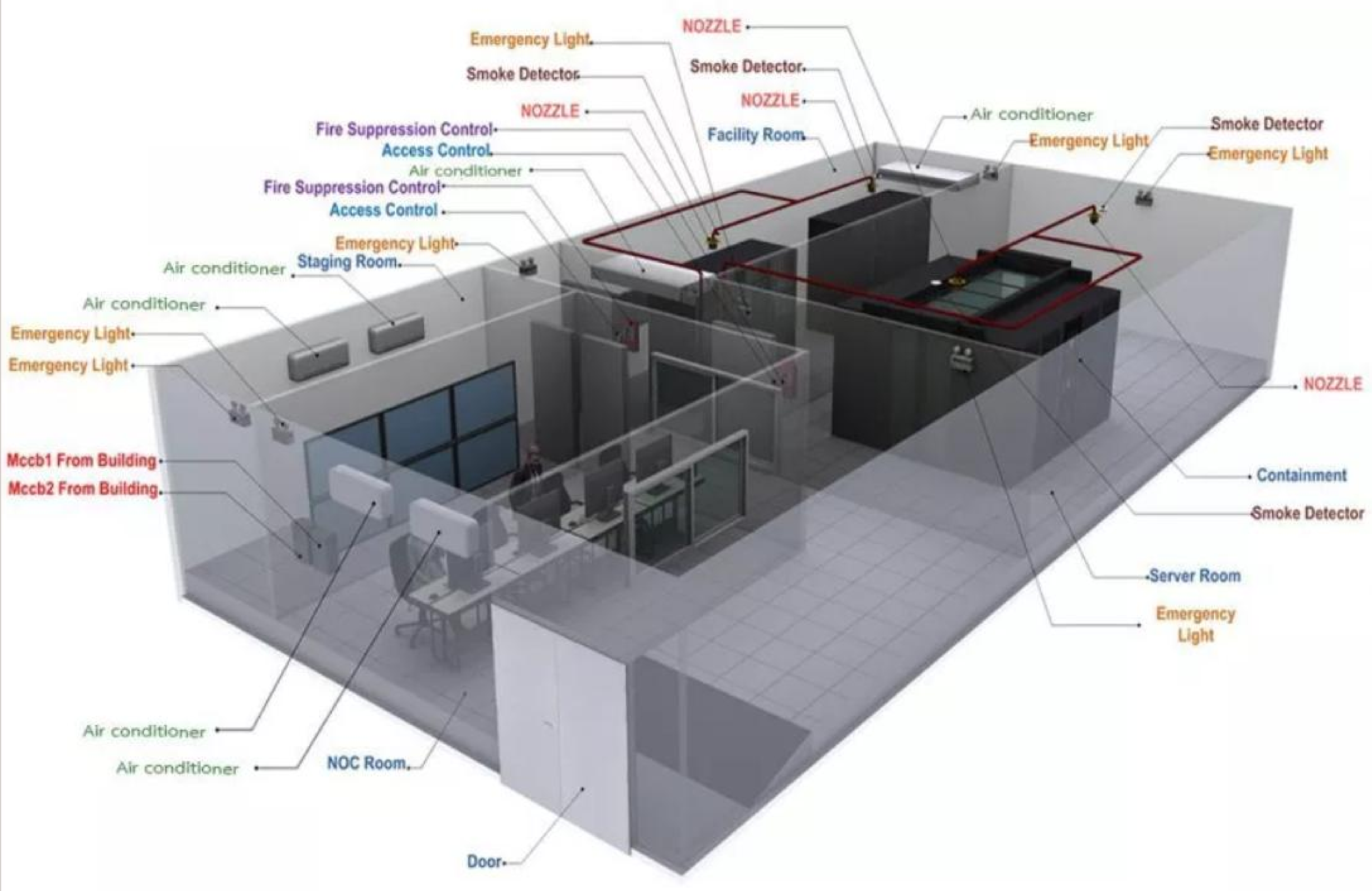
Homomorphic Tally

Using Paillier encryption, votes are tallied while encrypted. Election officials never see individual choices.

Anonymity Loop

Cryptographic nullifiers prevent double-voting without ever linking a voter's ID to their specific ballot.

How it Works



1. **Register:** Voter verifies identity and receives a PQC Keypair and unique Nullifier.

   Cast: Ballot is encrypted via Paillier and signed via Dilithium.

   Validate: System checks nullifier (double-vote check) and Dilithium signature (integrity check).

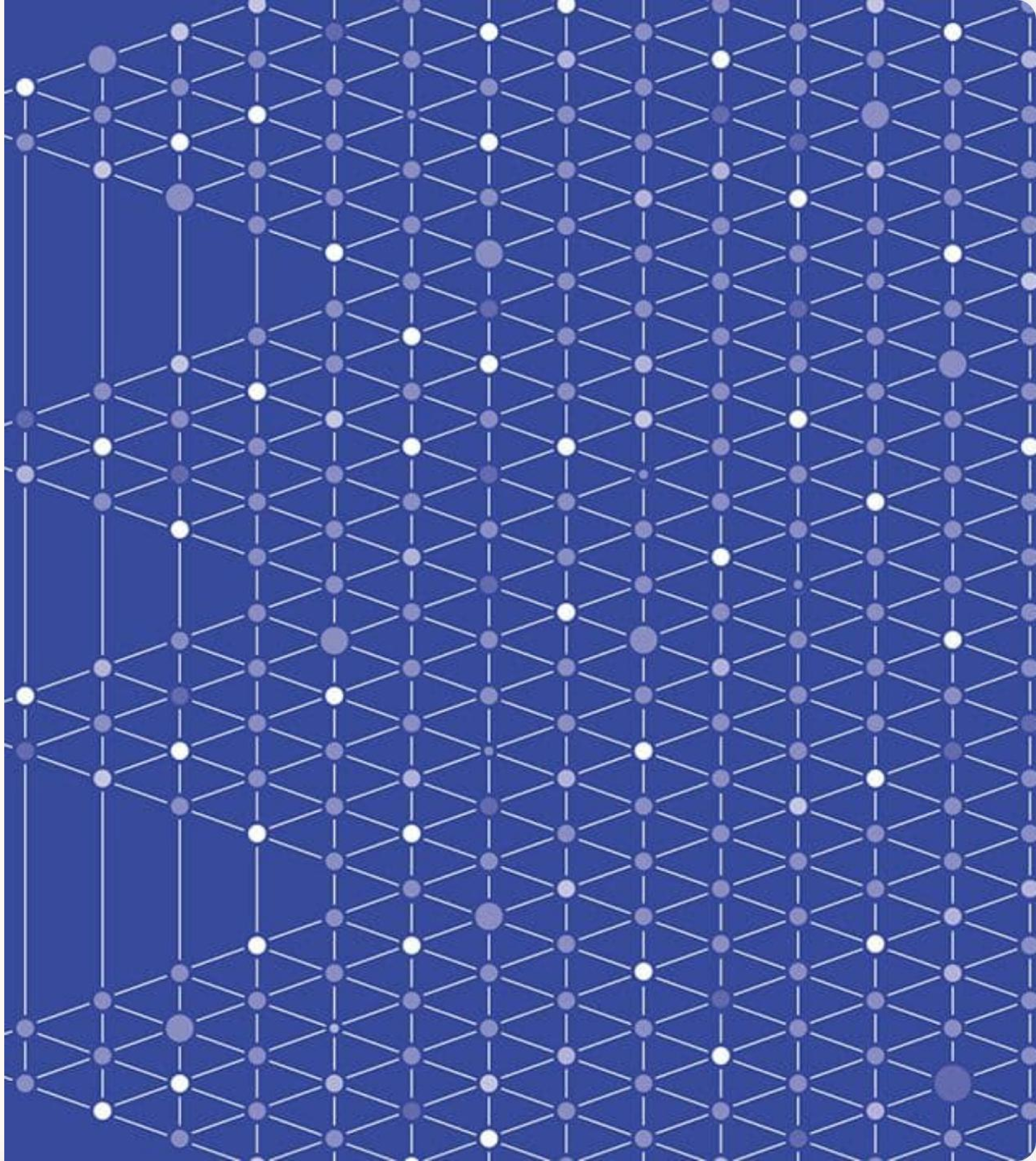
   Tally: Encrypted ballots are homomorphically summed. Private key decrypts ONLY the final result.

Lattice-Based Math

Current RSA/ECC relies on prime factorization—a problem Shor's algorithm solves effortlessly.

CRYSTALS-Dilithium uses "Lattices," a complex geometric structure that is mathematically hard even for quantum computers.

```
NIST Standard: FIPS-204
Security Level: Category 2/3/5
Resistance: Quantum + Classical
```



Zero-Knowledge Trust

0

LEAKED VOTES

Mathematically Forced Privacy

Our system doesn't rely on "admin ethics." It relies on the Paillier additive property:

$$Dec(Enc(V_1) + Enc(V_2) + ... + Enc(V_n)) = \Sigma V_i$$

The individual **Enc(V)** remains a black box. Only the **Sum** is revealed, guaranteeing end-to-end anonymity and verifiability.

Technology Stack

Component	Technology	Purpose
Backend	Python 3.x / FastAPI	High-performance, async API management
PQC Engine	liboqs (C/Python)	NIST Dilithium signature implementation
Privacy Layer	phe (Paillier)	Additive homomorphic encryption for tallying
Frontend	React / Tailwind	Authoritative, accessible user interface

Why Q-Vote?

Scale Neutral: Works for a 50-person club or a million-person city.

Audit-Ready: The "Bulletin Board" allows anyone to verify the tallying math without decrypting ballots.

Modular: Replace Dilithium with Falcon or Kyber as standards evolve.

The "Un-Hackable" USP

Most systems rely on a **Secure Database**. We rely on **Hard Math**. Even if the server is breached, the ballots remain encrypted and the privacy remains intact.

Roadmap to Democracy 2.0

Phase 1 (MVP)

Core Crypto + API + Generic ID Input

Phase 3 (Global)

Blockchain Audit Layer + Gov-Level Demo

Phase 2 (Scale)

Feasibility & Real-World Impact

This isn't just theory—it's a production-ready roadmap.

Low Overhead: Modern CPUs can sign Dilithium messages in milliseconds.

Ready Libraries: Leveraging *Open Quantum Safe* projects ensures we don't "roll our own crypto."

Immediate Relevance: With 2028-2029 elections looming, PQC-readiness is a national security mandate.

SECURE. ANONYMOUS. READY.

Thank You

Protecting the Ballot for the Quantum Era.

github.com/q-vote-demo   contact@qvote.tech

Image Sources

Thumbnail for freemindtronic.com

https://freemindtronic.com/wp-content/uploads/2024/10/quantum-computing-encryption-threats-freemindtronic-quantum-security-image-jpg.webp

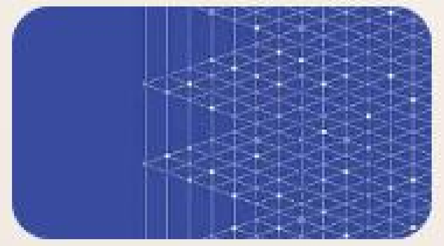
Source: freemindtronic.com

https://www.earth-changers.com/wp-content/uploads/2017/11/SDG-goals_16-peace-justice-strong-institutions-200x200-1.png

Source: www.earth-changers.com

https://canovate.com/wp-content/uploads/2024/08/Design-datacenter_%E0%B9%83%E0%B8%8A%E0%B9%89%E0%B8%88%E0%B8%A3%E0%B8%B4%E0%B8%87-1-e1645771697907.webp

Source: canovate.com



https://research.ibm.com/_next/image?url=https%3A%2F%2Fresearch-website-prod-cms-uploads.s3.us.cloud-object-storage.appdomain.cloud%2Fleadspace_post_quantum_telco_network_taskforce_9ef19c81c1.jpg&w=3840&q=75

Source: research.ibm.com

https://static.vecteezy.com/system/resources/thumbnails/014/290/345/small/team-of-engineer-architect-contractor-and-foreman-meeting-and-consulting-at-construction-building-site-with-floor-plan-for-real-estate-development-project-industry-and-housing-timeline-concept-free-photo.jpg

Source: www.vecteezy.com